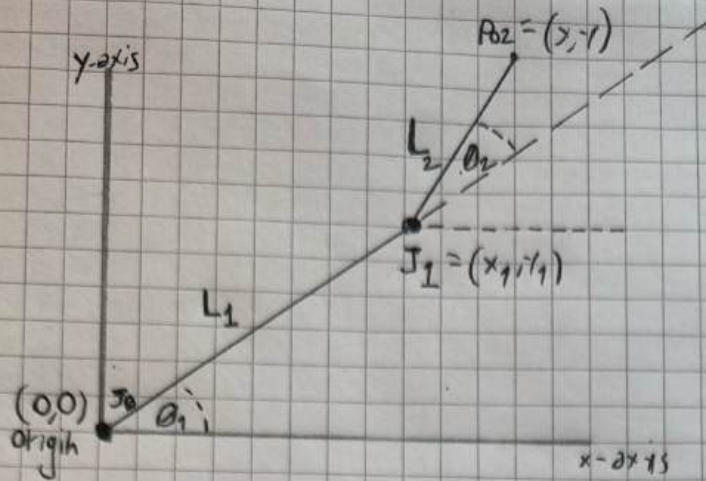




let L_1 be the length of link 1

let L_2 be the length of link 2

let θ_1 be the absolute angle of link 1 relative to the positive x-axis

let θ_2 be the relative angle of link 2, measured relative to the extension line of link 1

Therefore the absolute angle of link 2 relative to the positive x-axis is $(\theta_1 + \theta_2)$

By projecting link 1 onto the horizontal and vertical axes ~~by~~ using the right angle triangle formed with L_1 and θ_1

$$x_1 = L_1 \cos(\theta_1)$$

$$-y_1 = L_1 \sin(\theta_1)$$

By projecting link 2 onto the horizontal and vertical axes using its absolute angle $(\theta_1 + \theta_2)$

$$\Delta x = L_2 \cos(\theta_1 + \theta_2)$$

$$\Delta y = L_2 \sin(\theta_1 + \theta_2)$$

The end-effector's total position is the vector sum of the individual link projection

$$x = x_1 + \Delta x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)$$

$$y = -y_1 + \Delta y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2)$$

These equations are the complete forward kinematics map $f: (\theta_1, \theta_2) \rightarrow (x, y)$. Given any joint configuration, the end-effector will be unique.

Note:

This derivation extends to an arbitrary number of joints. For a planar kinematic chain of n links the absolute angle of link k is given by the sum of all preceding joint angles $\sum_{i=1}^k \theta_i$.

Consequently, the total position of the end-effector is found by summing the ~~pos~~ contribution of all individual links

$$x = \sum_{k=1}^n L_k \cos\left(\sum_{i=1}^k \theta_i\right)$$

$$y = \sum_{k=1}^n L_k \sin\left(\sum_{i=1}^k \theta_i\right)$$

This generalization holds in principle for the 7-joint KUKA iiwa, however the arm works in 3D space rather than a plane, requiring three-dimensional matrix transformation rather than simple planar trigonometry.